AWS Certified Developer Associate (Developing on AWS)

Course Overview

To earn the AWS Certified Developer – Associate certification, candidates must demonstrate proficiency in the following areas:

1. Understanding of core AWS services, uses, and basic AWS architecture best practices.
2. Proficiency in developing, deploying, and debugging cloud-based applications using AWS.
3. Understanding of the core AWS services, including knowledge of the AWS shared responsibility model, AWS CloudFormation, AWS Elastic Beanstalk, AWS CloudFront, AWS Lambda, AWS Cognito, AWS API Gateway, and more.
4. Proficiency in writing code for serverless applications.
5. Ability to use the AWS SDK to interact with AWS services and develop solutions.
6. Knowledge of application security best practices.

Key Features

- Our training modules have 50% - 60% hands-on lab sessions to encourage Thinking-Based Learning (TBL)
- Interactive-rich virtual and face-to-face classroom teaching to inculcate Problem-Based Learning (PBL)
- AWS certified instructor-led training and mentoring sessions provide a comprehensive learning experience tailored to individual needs, including sample questions for exam preparation.

- Well-structured use-cases to simulate challenges encountered in a Real-World environment
- Being an authorized AWS Training Partner gives us an edge over competition

Who should Enroll for this Course?

This course is intended for the following audience:

- Intermediate software developers

Prerequisites

- Participant should have attended AWS Technical Essential /Cloud Practitioner class/Basic knowledge of AWS is required.
- In-depth knowledge of at least one high-level programming language

Module 1 Course Overview

- Logistics
- Student resources
- Agenda
- Introductions

Module 2 Building a Web Application on AWS

- Discuss the architecture of the application you are going to build during this course
- Explore the AWS services needed to build your web application
- Discover how to store, manage, and host your web application

Module 3 Getting Started with Development on AWS

- Describe how to access AWS services programmatically
- List some programmatic patterns and how they provide efficiencies within AWS SDKs and AWS CLI
- Explain the value of AWS Cloud9

Module 4 Getting Started with Permissions

- Review AWS Identity and Access Management (IAM) features and components permissions to support a development environment
- Demonstrate how to test AWS IAM permissions
- Configure your IDEs and SDKs to support a development environment
- Demonstrate accessing AWS services using SDKs and AWS Cloud9
- Lab 1: Configure the Developer Environment
- Connect to a developer environment
- Verify that the IDE and the AWS CLI are installed and configured to use the application profile
- Verify that the necessary permissions have been granted to run AWS CLI commands
- Assign an AWS IAM policy to a role to delete an Amazon S3 bucket

Module 5 Getting Started with Storage

- Describe the basic concepts of Amazon S3
- List the options for securing data using Amazon S3
- Define SDK dependencies for your code
- Explain how to connect to the Amazon S3 service
- Describe request and response objects

Module 6 Processing Your Storage Operations

- Perform key bucket and object operations
- Explain how to handle multiple and large objects
- Create and configure an Amazon S3 bucket to host a static website
- Grant temporary access to your objects
- Demonstrate performing Amazon S3 operations using SDKs
- Lab 2: Develop Solutions Using Amazon S3
- Interact with Amazon S3 programmatically using AWS SDKs and the AWS CLI
- Create a bucket using waiters and verify service exceptions codes
- Build the needed requests to upload an Amazon S3 object with metadata attached
- Build requests to download an object from the bucket, process data, and upload the object back to the bucket
- Configure a bucket to host the website and sync the source files using the AWS CLI
- Add IAM bucket policies to access the S3 website.

Module 7 Getting Started with Databases

- Describe the key components of DynamoDB
- Explain how to connect to DynamoDB
- Describe how to build a request object
- Explain how to read a response object
- List the most common troubleshooting exceptions

Module 8 Processing Your Database Operations

- Develop programs to interact with DynamoDB using AWS SDKs
- Perform CRUD operations to access tables, indexes, and data
- Describe developer best practices when accessing DynamoDB
- Review caching options for DynamoDB to improve performance
- Perform DynamoDB operations using SDK
- Lab 3: Develop Solutions Using Amazon DynamoDB
- Interact with Amazon DynamoDB programmatically using low-level, document, and high- level APIs in your programs
- Retrieve items from a table using key attributes, filters, expressions, and paginations
- Load a table by reading JSON objects from a file
- Search items from a table based on key attributes, filters, expressions, and paginations
- Update items by adding new attributes and changing data conditionally
- Access DynamoDB data using PartiQL and object-persistence models where applicable

Module 9 Processing Your Application Logic

- Develop a Lambda function using SDKs
- Configure triggers and permissions for Lambda functions
- Test, deploy, and monitor Lambda functions
- Lab 4: Develop Solutions Using AWS Lambda Functions
- Create AWS Lambda functions and interact programmatically using AWS SDKs and AWS CLI
- Configure AWS Lambda functions to use the environment variables and to integrate with other services
- Generate Amazon S3 pre-signed URLs using AWS SDKs and verify the access to bucket objects

- Deploy the AWS Lambda functions with .zip file archives through your IDE and test as needed
- Invoke AWS Lambda functions using the AWS Console and AWS CLI

Module 10 Managing the APIs

- Describe the key components of API Gateway
- Develop API Gateway resources to integrate with AWS services
- Configure API request and response calls for your application endpoints
- Test API resources and deploy your application API endpoint
- Demonstrate creating API Gateway resources to interact with your application APIs
- Lab 5: Develop Solutions Using Amazon API Gateway
- Create RESTful API Gateway resources and configure CORS for your application
- Integrate API methods with AWS Lambda functions to process application data
- Configure mapping templates to transform the pass-through data during method integration
- Create a request model for API methods to ensure that the pass-through data format complies with application rules
- Deploy the API Gateway to a stage and validate the results using the API endpoint

Module 11 Building a Modern Application

- Describe the challenges with traditional architectures
- Describe the microservice architecture and benefits
- Explain various approaches for designing microservice applications
- Explain steps involved in decoupling monolithic applications
- Demonstrate the orchestration of Lambda Functions using AWS Step Functions

Module 12 Granting Access to Your Application Users

- Analyze the evolution of security protocols
- Explore the authentication process using Amazon Cognito
- Manage user access and authorize serverless APIs
- Observe best practices for implementing Amazon Cognito
- Demonstrate the integration of Amazon Cognito and review JWT tokens

- Lab 6: Capstone – Complete the Application Build
- Create a Userpool and an Application Client for your web application using
- Add new users and confirm their ability to sign-in using the Amazon Cognito CLI
- Configure API Gateway methods to use Amazon Cognito as an authorizer
- Verify JWT authentication tokens are generated during API Gateway calls
- Develop API Gateway resources rapidly using a Swagger importing strategy
- Set up your web application frontend to use Amazon Cognito and API Gateway configurations and verify the entire application functionality

Module 13 Deploying Your Application

- Identify risks associated with traditional software development practices
- Understand DevOps methodology
- Configure an AWS SAM template to deploy a serverless application
- Describe various application deployment strategies
- Demonstrate deploying a serverless application using AWS SAM

Module 14 Observing Your Application

- Differentiate between monitoring and observability
- Evaluate why observability is necessary in modern development and key components
- Understand CloudWatch's part in configuring the observability
- Demonstrate using CloudWatch Application Insights to monitor applications
- Demonstrate using X-Ray to debug your applications
- Lab 7: Observe the Application Using AWS X-Ray
- Instrument your application code to use AWS X-Ray capabilities
- Enable your application deployment package to generate logs
- Understand the key components of an AWS SAM template and deploy your application
- Create AWS X-Ray service maps to observe end-to-end processing behavior of your application
- Analyze and debug application issues using AWS X-Ray traces and annotations

Module 15	Course Wrap-up

- Course overview
- AWS training courses
- Certifications
- Course feedback

About CloudThat

CloudThat is the first company in India to offer Cloud Training & Consulting services for mid-market & enterprise clients from across the globe. Since our inception in 2012, we have trained over 500K IT professionals from fortune 500 companies on technologies such as Microsoft Azure, Amazon Web Services, VMware, Artificial Intelligence, Machine Learning, Google Cloud, IoT, OpenStack, OpenShift, DevOps, Big Data, Kubernetes, and more.

With an expertise in major Cloud platforms, CloudThat is a proud Microsoft Gold Partner, AWS Authorized Training Partner, VMware Authorized Training Reseller, and Google Cloud Platform Partner. Till date we have conducted successful corporate training for some of the fortune 500 companies which includes Accenture, Deloitte, Infosys, Fidelity, HCL, Intuit, GE, TCS, HP, SAP, Oracle, Western Union, Philips, Flipkart, L&T and Samsung, just to name a few. We have presence in Bengaluru, Mumbai, USA & UK, offering on-site and pre-scheduled public batches in different IT centric cities of India and Overseas. Overseas.

Our Success Track

12+ Years of Experience	**200+** Corporates served	**30+** countries catered
750K+ Professionals trained	**350+** Projects delivered	**600+** Cloud certifications

To know more about our cloud certification training, email at **sales@cloudthat.com** or call us at **+918880002200.**